Can AI Agents Be Organized More Efficiently for Discovering Science?

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Abstract

Pruning and quantization shrink neural networks because large networks are redundant: many channels carry the same signal, so removing them costs little. We apply the same tools to teams of language-model agents and find the same redundancy, with a sharper consequence for anyone who pays for them. A team can spend its budget on *width* (more calls whose answers are voted on, like the channels pruning removes), on *depth* (more reasoning per call, like the bits quantization removes) or on *checking* (a verifier that picks one answer, like re-ranking). The members of a team fail on the same questions. Six copies of one model carry the information of 1.9 independent answers, so a 19-agent team can be pruned heavily with a loss inside the noise, and it does not beat one call to a stronger model. Checking selects but cannot create: on the 20 questions no member answered, critics and refiners recovered none. At equal dollars, one step of extra reasoning beat a vote of quick copies in a preregistered test on sealed questions (by 3.7 and 10.7 points for two cheap models). The knob that moved accuracy per dollar most was the choice of member. On LAB-Bench biology questions one call to a strong model matched every team at a fraction of the best team's cost, and when a team building classifiers on physics and ecology tables lost a member, the planner model, not the recovery rule, set the result. A factor model with one difficulty per question prices this shared failure on public leaderboards, and a replay tool picks the cheapest members for a budget from calls already made.

1. Introduction

A biologist asks an AI assistant why most cells died in an amnion-isolation protocol, and the answer is wrong. Four knobs are within reach: ask again, ask the model to think harder, ask a second model to check, or switch to a stronger model. Each costs money, and each changes the system's behavior in its own way. This paper measures what each knob buys.

The question has a familiar shape for anyone who compresses neural networks. Compression asks which parts of a network carry information and which are redundant: pruning removes units whose signal other units already carry (Han et al., 2016; He et al., 2018), and quantization lowers precision until accuracy starts to fall (Kumar et al., 2024). A team of agents is a network whose units are model calls. Asking again adds width, an ensemble of copies; thinking harder or switching models adds depth, the counterpart of precision; a checking model is re-ranking. We took pruning, quantization and architecture search to such teams (Figure 1) and found the redundancy that compression relies on. Its consequence for a budget is the subject of this paper: *accuracy per dollar is set by which members a team calls, not by how it wires them.*

One question, followed through a team. Our first team had six proposers, six critics, six refiners and an aggregator, all copies of one small model, working on hard reasoning puzzles from BIG-Bench Extra Hard (BBEH; Kazemi et al., 2025). One puzzle lists dozens of statements such as "Mindy says Olga lies" and asks which of three people tell the truth (Table 1). Only 1 of the 6 proposers answered it correctly. The other 5 gave 5 different wrong answers, so a vote had nothing to work with. Then 4 of the 6 critics re-derived the right answer from the one correct proposal, all 6 refiners agreed, and the team answered correctly. This is a team at its best: it found the one right answer and amplified it. The other side is just as clear. On the 20 questions where no proposer was right, the team recovered none (0.0%). Critics and refiners select among answers someone produced; they do not produce new ones, just as no later layer can restore information that no channel carries.

Copies are correlated channels. The example generalizes because copies fail together. The six proposers' right-or-wrong records correlate at $\rho = 0.43$, so by the design effect of survey sampling (Kish, 1965) they are worth

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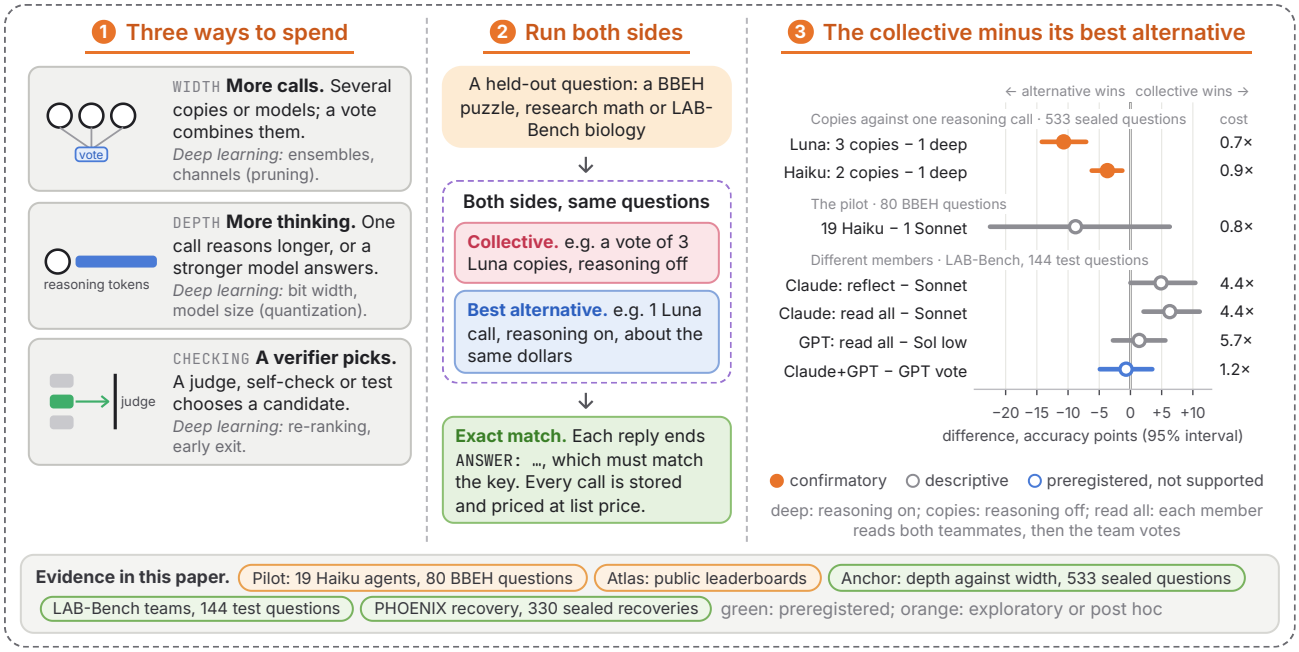


Figure 1. How every comparison in this paper works, and what it found. (1) A budget can buy width (more calls whose answers are combined), depth (more reasoning per call, or a stronger model) or checking (a verifier choosing among candidates); each has a deep-learning counterpart. (2) A collective and its best single alternative answer the same held-out questions, every reply is graded by exact match with the answer key, and every call is stored and priced. (3) The collective minus its best alternative, in accuracy points with 95% paired-bootstrap intervals; left of zero the alternative wins, and the right column is the collective's cost as a multiple of the alternative's. Rows 1–2 (CONFIRMATORY; 533 sealed questions): a vote of reasoning-off copies against one reasoning-on call of the same model at about the same dollars. Row 3: the pilot's 19-agent Haiku team against one Sonnet call (80 BBEH questions). Rows 4–7 (LAB-Bench, 144 test questions): a team of a small and a mid-sized model (Claude) with every member thinking again, or reading both teammates, against its best member alone; a team built around a strong model (GPT) against that model alone; and a preregistered panel mixing the two families against one family's own vote. The box below lists the studies; green marks those with preregistered tests.

$6/(1+5\rho) = 1.9$ independent answers. The chance that at least one of the six is right reaches 75.0%, against 97% if their errors were independent, and a plurality vote rises only from 44.6% to 52.1% (Figure 2). This is the textbook failure condition for ensembles: the Condorcet jury theorem needs independent voters (Ladha, 1992), and an ensemble improves on its members only through their disagreement (Krogh & Vedelsby, 1995). Redundancy is also why the team prunes well: 19 of its 27 pruned structures, the full team included, lie inside the full team's 95% interval of ±11 points. And it is why the full team did not beat one call to a stronger model, whose lead was +8.8 points (−6.3 to +22.5).

Turning one knob changes the behavior. Interpretability research shows that turning up a single internal feature of a model, such as one that represents the Golden Gate Bridge, changes what the model says in a predictable direction (Templeton et al., 2024). A team has a handful of such knobs on the outside, and we turned each with the others held fixed (Figure 1). *Width* moves coverage slowly and the vote hardly at all. *Depth* moves accuracy: on 533 sealed questions, raising a cheap model's reasoning by one step beat a vote of its quick copies at about the same dollars, by 3.7 and 10.7 points for two models (CONFIRMATORY). On public leaderboards the higher setting shifts a model's ability without changing which questions are hard (Section 4), much as precision moves a network's effective parameter count in a precision scaling law (Kumar et al., 2024). *Checking* is capped by what the members cover. *The member* moves the most. On LAB-Bench biology (Laurent et al., 2024), a team of a small and a mid-sized model gained +6.3 points over its best member when every member read the others' answers, at 4.4× the cost, yet one call to a stronger model matched every team at 1/8 of the cost of the best team of smaller models. In PHOENIX, a three-agent team building classifiers on physics and ecology tables loses a member mid-run; the planner model set the quality of the recovery, and a smarter recovery rule added no gain we can credit to the rule.

Why this matters to people who use these systems. For a scientist who uses an assistant every day, the knobs translate into rules of thumb. Asking again mostly repeats the mistake, and five agreeing answers are not five confirmations. One step of extra reasoning is the cheapest gain. A

checker helps only when some answer is already right. A stronger model helps most, and can cost less than a team of weaker ones. The behavior worth regulating is shared failure: the questions every member gets wrong are invisible to votes and to agreement scores, and agreement-based selection can even rate a weaker model above the strongest one (Section 6). We give a factor model that measures shared failure from graded questions, and a tool that turns stored calls into a choice of members for each budget (Section 8).

Data and outline. Table 1 shows one item from each source and how it is graded: BBEH puzzles and recent research mathematics for the preregistered tests of depth against width, LAB-Bench DNA-sequence and protocol questions for the team studies, three public science tables for PHOENIX, and public leaderboards for the factor model. Every model call is stored with its price, so every team in this paper is rescored by replay at no cost. Section 3 follows the pilot, Section 4 measures shared failure on public data, Sections 5 and 6 test depth and checking, Section 7 changes the members and the coordination rules, and Section 8 turns the results into a budget map.

Contributions.

- **A knob-by-knob measurement** of width, depth, checking and member choice at matched dollars, with preregistered tests read once on sealed questions.
- **The Atlas**: a Rasch-type factor model (Rasch, 1960) with one difficulty per question, fitted to 173 SWE-bench Verified submissions, 126 MathArena configurations and 165 LiveBench models. It prices the questions that every held-out system fails better than independence or a one-factor copula.
- **Team studies** on biology questions and science tables: a preregistered rule for when a member should ask a teammate, a preregistered panel mixing two model families, and PHOENIX, a recovery rule for a lost member, scored on 330 sealed recoveries.
- **A budget map** that picks the best members for each dollar and latency budget from stored calls, and a process record: 135 independent audits and $196.64 of API spend, with every number in the paper generated from a committed file and labelled with its status (Appendix A).

2. Definitions

We define each term in words first. **Width**: more calls of the same setting, or of different systems, combined by a rule (a plurality vote) or a selector. **Depth**: more reasoning per call (a higher effort or thinking budget) or a stronger tier. **Verification**: choosing among candidate answers, by a judge model, a self-check, execution, or a label-free rule such as Dawid–Skene (Dawid & Skene, 1979).

The factor model. System s answers item i correctly with probability $P_{si} = \sigma(a_s - \beta_s b_i)$, $\sigma(x) = 1/(1 + e^{-x})$. The *ability* a_s is how good the system is overall; the *difficulty* b_i is how hard the item is for everyone, the shared part of failure (scaled to mean 0 and standard deviation 1); the *failure slope* β_s is how sharply the system's success falls with difficulty. Level L0 fixes $\beta_s = 1$. A *family shock* $u_{f(s),i}$ adds a shared draw for systems of one vendor; *call luck* is what remains, identified only from repeated calls. Fits are penalised logistic regressions (ridge 0.1), so items every system gets right or wrong keep finite difficulties.

Derived quantities. The *systematic share* $\mathrm{var}(b)/(\mathrm{var}(b) + \pi^2/3)$ is the share of latent variation in success that items share ($\pi^2/3$ is the variance of the logistic call-level noise). *Coverage* of a set of systems is the share of items at least one member gets right: the ceiling of any rule that returns one member's answer, as Chen (2026) also show; the *all-wrong rate* is one minus coverage. The *vote ceiling* of one system is the share of items whose most common answer is right, the limit of infinitely many repeats. *Selection efficiency* $\eta = (\text{selected} - \text{best single})/(\text{coverage} - \text{best single})$ is the share of the room above the best member that a selector reaches. A *consensus inversion* is a label-free rule rating a weaker system above the best one. *Calibration by dilution*: adding near-random votes shrinks a panel's calibration gap without improving its ranking (F13 in our findings ledger).

Statuses. CONFIRMATORY: a preregistered test in a Holm family, read once on sealed items. PREREGISTERED-DESCRIPTIVE: a preregistered quantity, read once, no pass/fail. EXPLORATORY: on sealed items, not preregistered as a test. POST HOC: chosen after the data were read, or on data not sealed for it; all public-data analyses are post hoc in this sense, as are numbers computed after a read. Post hoc results are many and uncorrected for multiplicity; we read them as descriptions, not tests.

3. The pilot: nineteen agents and one call

Our first study pruned a team rather than measuring a lever (Figure 1, row 3). Six Claude Haiku 5.5 proposers answered each question, six critics read all six answers, six refiners revised, and one aggregator answered: 19 calls. On 80 BBEH questions the full team scored 62.5% at $0.0115 per question; one Sonnet 5.5 call with thinking off scored 71.3% at $0.0144, a difference of +8.8 points (−6.3 to +22.5; EXPLORATORY). The pruned teams tied within this noise.

Table 1. The data behind each finding, with one item from each source (abridged: long passages elided with . . .). Every reply ends with an answer line that is graded against the key; the science tables are scored on a sealed test split.

Source and use	One item	Graded by
BBEH puzzles (Kazemi et al., 2025): the pilot (80 questions) and the anchor (440 sealed)	*Web of lies.* "Assume each person either always tells the truth or always lies. Mindy says Olga lies. Mustafa says Odell lies. . . . Do Vina, Melanie, and Helene tell the truth?" Key: `no, unknown, yes`	Exact match of the answer line, after case and punctuation are normalised
Research mathematics (MathArena; Balunović et al., 2025): the anchor (93 sealed)	A final-answer problem taken from a recent arXiv paper, with one closed-form answer such as a number or an expression.	Symbolic equivalence with the key
LAB-Bench SeqQA (Laurent et al., 2024): the team studies (90 test questions)	"What is the expected amplicon length if I used the primer pair TCAAGCTCAACCACAGGACCC, GATGGCAGCATCTTGATCTACT in a PCR from the following template: ATGGCCAACCC. . . ?" Options A. 439, B. 470, C. 596, D. 838, E. insufficient information. Key: C	Exact option letter
LAB-Bench ProtocolQA: the team studies (54 test questions)	A multi-step protocol for isolating amnion cells, then the outcome "the vast majority of the cells are dead" and a question asking which step was done wrong. Option B: "I omitted in the step 1d to pre-wet the second sterile tray with washing buffer with antibiotics." Key: B	Exact option letter
Science tables: PHOENIX (18 cases)	One row of HIGGS (Baldi et al., 2014): 28 kinematic measurements of a simulated collision, labelled signal or background. Forest cover type (Blackard & Dean, 1999): 54 cartographic features of a plot, labelled with one of 7 tree covers. MiniBooNE (Roe et al., 2005): 50 particle-identification features, labelled signal or background.	Balanced accuracy of the recovered classifier on a sealed test split, scored once
Public leaderboards: the Atlas	SWE-bench Verified (Jimenez et al., 2024): 173 submissions on 500 GitHub issues, each resolved or not; MathArena (Balunović et al., 2025) and LiveBench (White et al., 2024).	Each leaderboard's own grading

Across the 80 questions the worked example of Section 1 is the rule (Figure 2). Accuracy rose through critique (57.5%) and refinement (60.2%) to 62.5%; on the 19 questions where one or two proposers were right the team reached 73.7%, and on the 20 where none was right 0.0%. The effective size $n/(1+(n-1)\rho)$ measures redundancy, not a number of agents one could deploy, and the stages were not ablated one by one.

4. The Atlas: what public data say about width and depth

Data. SWE-bench Verified (Jimenez et al., 2024): 173 submissions on 500 tasks, one run each, 39 with per-task cost and step counts (mini-SWE-agent). MathArena (Balunović et al., 2025): 126 configurations on 733 final-answer problems from 23 competitions, usually four calls each, with answers grouped into classes by the grader. LiveBench (White et al., 2024): 165 models on 228 items of three binary tasks, one call each, without answers or costs. Each benchmark is fitted on its own. All Atlas numbers are POST HOC.

4.1. Failures are mostly shared

The systematic share is 0.73 on SWE-bench Verified (item bootstrap 0.70–0.75; over 40 random halves of the systems, 0.71–0.73), 0.73 on MathArena (0.72–0.75) and 0.65 on LiveBench (0.61–0.68); our Phase 1 BBEH panel of four arms sits lower at 0.42 (0.38–0.47; Appendix Figure 6). On the 39 SWE-bench submissions with per-task data, difficulty alone implies 98% of the mean pairwise failure correlation (0.502 of 0.512); two systems of one company add only 0.026 of residual correlation. All 39 fail 8.6% of tasks, against 0.0% if they failed independently.

4.2. Width is capped by the shared tail

Coverage and the all-wrong tail. Fitting difficulties on half the systems and each held-out system's ability on half the tasks, the model predicts the coverage of held-out 2–5 system sets on held-out SWE-bench tasks within 1.5–1.8 points; independence misses by 11–20. Chen (2026) report that a calibrated one-factor copula underprices the all-wrong tail by about 2.5× on open-ended mathematics. We give the factor model and the copula the same information, the calibration tasks only: the factor model draws an unknown test task's difficulty from the calibration tasks' difficulties, the copula fits its correlation to the calibration tasks' pairwise failures. For all 87 held-out SWE-bench systems on 250 held-out tasks the observed all-wrong rate is 7.6% (Clopper–Pearson 4.6–11.6); the factor model predicts 5.9% (observed over predicted 1.29, 0.68–1.84), the copula 3.5% (2.15×) and independence 0.00% (Figure 9). The factor model's free distribution of difficulties carries the tail that a Gaussian latent factor misses. On the small MathArena sets (15–26 test problems) both models underpredict the 5-system tail,

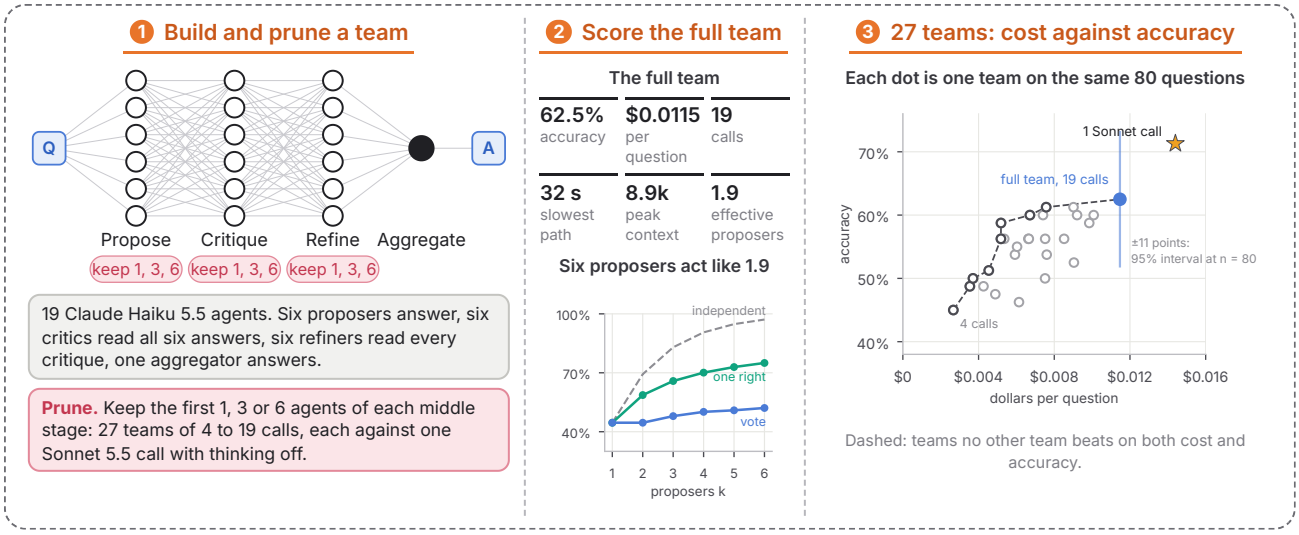


Figure 2. The pilot. (1) Nineteen Claude Haiku 5.5 agents in four stages; pruning keeps the first 1, 3 or 6 agents of each middle stage, which gives 27 teams. (2) The full team's measurements, and why it does not gain more: a plurality vote of k proposers (averaged over every set of k) against the share of questions at least one of the k answered right, and that share if their errors were independent; six proposers are worth 1.9 independent answers. (3) All 27 teams on the same 80 BBEH questions, accuracy against dollars per question; the vertical bar is the full team's 95% interval, wide enough to cover most teams, and the star is one Sonnet 5.5 call with thinking off (EXPLORATORY).

on 2 and 2 of 6 sets; the math tail is unresolved.

Votes. The factor model's vote rule ranks where votes of one system pay but overstates the gains, and votes pay most when weak systems with different strengths are mixed, where the model prices the composition without bias (Appendix B).

4.3. Depth moves ability, not the failure pattern

For each of 19 public pairs of one model at two adjacent reasoning settings (11 models, some on two benchmarks; difficulties from every other system), we fit both settings' ability and slope on the items both answered (Figure 10). Pooled with random effects, ability rises by 0.92 logits (0.59–1.24; between-pair variance 0.35); the slope changes by 0.03 (−0.14–0.21; between-pair variance 0.00). One pair per model and benchmark (13 pairs) gives 1.21 and 0.11. Held out, the higher setting's calls are predicted better by the lower setting plus one shift than by a refit that may also change the slope (log loss per call 0.0041 in favour of the shift, 0.0022–0.0061) and far better than by its base rate (0.192 nats per call). The difficulties are the population's, so this says the higher setting fails the items that are hard for everyone; the lower setting's own item outcomes plus a shift predict no better (0.007, −0.017 to 0.030). Two pairs left out after a look at output tokens (one setting of unknown order, one different checkpoint) are shown in the extension report and were never pooled.

Depth against width at equal dollars (post hoc). For 10 MathArena pairs whose lower setting has three or more calls, one call at the higher setting minus the vote of the lower setting with the nearest dollars per item (the 3-vote in every pair) is 12.7 points (7.1–18.2); on the 7 pairs priced within a factor of 1.5 it is 11.9 (6.0–17.7; Appendix Figure 7). The step pays where a model starts low and about nothing at the top of its range: +1.1 points for Gemini 3.1 Pro medium to default and −1.5 for Opus 5.5 high to max. We logged, before it is read, that this predicts about zero for the preregistered CD2 test of a top Haiku effort step. Price matters as much: on SWE-bench Verified MiniMax M2.5 (high) resolves 75.8% at $0.073 per task against 76.8% at $0.754 for Claude Opus 4.5 (high), 10× the price for one point (Figure 9, right).

5. Preregistered API studies

Phase 1: the saturated case. On 300 sealed BBEH items (Kazemi et al., 2025), one GPT-6.1 Sol call at low reasoning was right 97.3% of the time at $0.0062 per call; Claude Sonnet 5.5 without reasoning 72.0% at $0.0139 (PREREGISTERED-DESCRIPTIVE). Width bought little: 5 Haiku calls are worth 1.84 independent calls (repeat correlation 0.43), their plurality reached 48.3%, a 12-call mixed panel 58.0%, and none of 276 cascade settings came within 2 points of Sol at 0.8× its dollars (EXPLORATORY/POST HOC). The confirmatory cascade test (H7) passed non-inferiority to Sonnet (−0.7 points) only by escalating 78% of items. Sol's accuracy is not answer memory: 40 of 40 perturbed items

right. The Sol–Sonnet gap mixed vendor and reasoning, which motivated the anchor.

The anchor. 440 BBEH items (linguistic puzzles, cartoon captions, word sorting, causal judgement) and 93 arXiv-Math problems were sealed before any call. Arms: Claude Haiku 5.5 (off, low, high), Sonnet 5.5 (off, low), GPT-6 Luna (none, low, medium), GPT-6.1 Sol (low, medium), most with three calls per item; dollars are measured at list prices. Each block was read once with frozen code; Holm at 5% over the two tests that kept power of at least 0.70 (AH2, AH3).

*AH2 (*CONFIRMATORY*, passes; $p = 0.001$).* Raising a cheap model's reasoning one step beat repeating its no-reasoning call at the nearest measured dollars: Haiku at effort low against the Haiku-off 2-vote by 3.7 points (1.3–6.2), Luna at reasoning low against the Luna-none 3-vote by 10.7 (7.1–14.2); the deeper call cost 1.17× and 1.38× as much. For Haiku the 2-vote equals one call in expectation, so this shows that effort low beats effort off at slightly more dollars, not that depth beats a wider vote; against the dearer 3-vote the difference is 1.8 (−0.7–4.3), not tested. The effect sits mainly in word sorting (and linguistic puzzles for Luna; computed after the read). The decision as frozen: for cheap models on these items, buy the first effort step before width. It covers neither higher steps nor other tasks.

*AH3, the vote rule (*CONFIRMATORY*, passes).* Cross-fitted on single calls, the factor model predicted the six 3-call arms' 3-vote accuracy with mean absolute error 0.8 points (upper bound 2.4) and their gain over one call within 0.7 (upper bound 1.3). The gains were small (0.4 to 2.6 points) and zero-gain also meets the limits (gain error 1.5 against 0.8), so AH3 shows accuracy where votes gain little; P1 above finds the same rule overstating larger gains.

The depth study. A preregistered study of Haiku at every reasoning effort (research/v4/cd/read.md) finds that accuracy gains flatten by the second-highest effort; the drop at the highest effort is within its interval and coincides with calls cut at the output limit.

6. Verification

Label-free selection. With repeats, Dawid–Skene rated Sol at 62.5% on Phase 1 against a true 97.3% (EXPLORATORY); on the anchor its pooled variant put Sonnet at 79% against a true 45%, and its answers (47.5%) stayed below one Sol low call (56.2%). A simulation map (Figure 11) shows when this happens: with one strong system among weak ones and one call each, plain Dawid–Skene ranks a weak system first in 90% of panels; a fixed label space removes the inversion in the simulation (0%) but still loses 8.0 points to the strong system. On MathArena the fixed label space does not help: beside a lone strong system it ranks another system first in 62–71% of panels, against 42–63% for plain Dawid–Skene. At matched calls (three systems with one call each, against the best system's own 3-vote, the best chosen with labels) every policy loses beside a lone strong system (plurality −34.0 points, plain Dawid–Skene −25.7) and none gains among competent ones (plurality −1.8, plain Dawid–Skene −1.0, fixed labels −0.3).

The anchor's judge, and why it cannot test judges. On the anchor's BBEH items the Sol low judge, choosing among the answers of two cheap pools, reached 52.3%: 13.3 points above the best pool arm (9.8–15.8), selection efficiency 0.70 (0.57–0.80). Sol low answering alone scored 60.6%, so the judge sat 8.3 points below it (−12.0 to −4.8) at 1.6× the dollars (Appendix Figure 8; POST HOC, computed after the read). But the pools covered only 58.0% of the items: even a perfect selector over them would have lost to Sol alone. The comparison measures the pools, not the judge's signal; a test of judges needs pools whose coverage exceeds the judge model's own accuracy. On math the judge was capped by 22.6% coverage.

7. Different members and coordination rules

Different members. If copies fail together, members that fail differently should help (Figure 3). On 144 held-out LAB-Bench questions (DNA-sequence manipulation and protocol troubleshooting), each provider fielded one cheap model and one strong model at two reasoning efforts: Claude Haiku 5.5 with Sonnet 5.5 at low and medium effort (two models, three settings), and GPT-6 Luna with GPT-6.1 Sol at low and medium. With every member reading both teammates' answers before revising, the Claude team scored 84.7% against 78.5% for Sonnet low alone, +6.3 points (+2.1 to +11.1; POST HOC) at 4.4× the cost; a plain vote of the three first answers gained +2.1 (−0.7 to +5.6). Sol low alone scored 86.8% at $0.006 per question, and no GPT team beat it by a resolved margin (every member reading both teammates: +1.4, −2.8 to +5.6). The best Claude team (85.4% at $0.043) stayed below that one Sol call. A preregistered panel mixing the providers, chosen on practice questions and replayed from stored calls at no cost, scored 86.8% at $0.023: −0.7 points against GPT's own vote (−4.9 to +3.5) and +6.3 against Claude's (+0.7 to +11.8). The claim needed both to be positive and is not supported (CONFIRMATORY). The two providers refused disjoint questions, too few to matter for a vote.

A learned rule for when to ask a teammate. After its first answer each member could stop, think again alone, or ask one teammate. A per-member rule was fitted on practice questions, frozen, and read once; it had to be cheaper

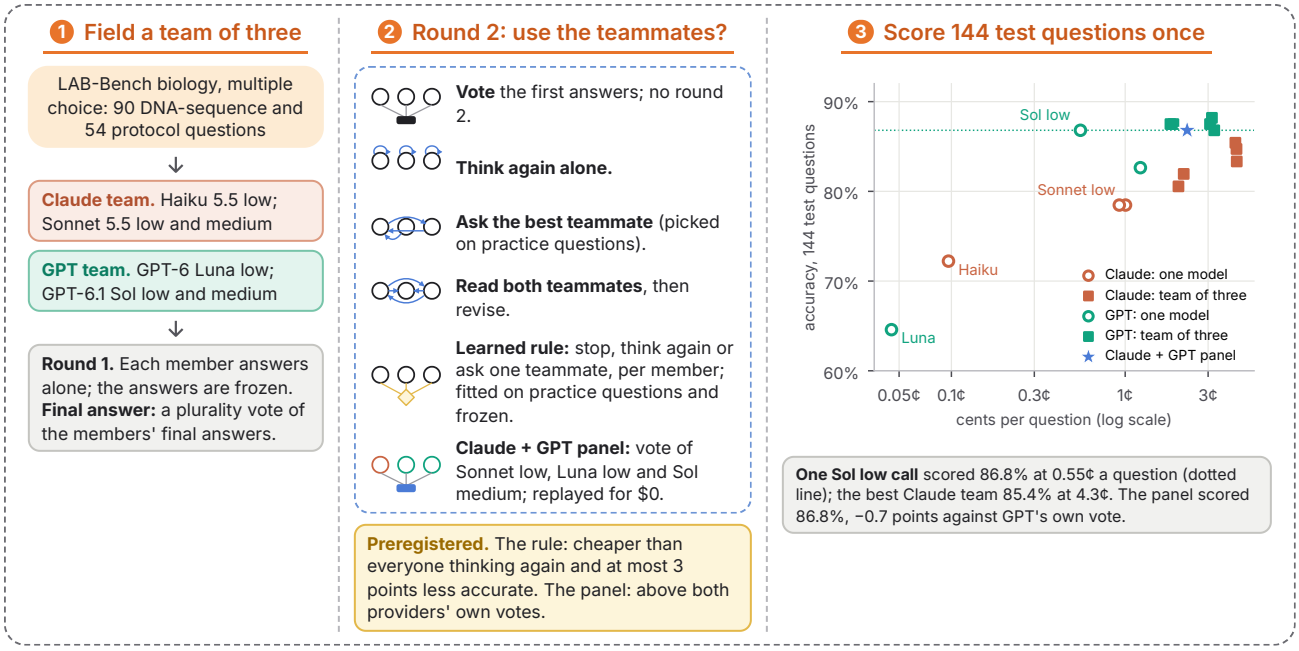


Figure 3. The LAB-Bench team study. (1) Each provider fields one cheap model and one strong model at two reasoning efforts; every member answers alone first, and the team's answer is a plurality vote of its members' final answers. (2) The round-2 methods compared: a plain vote, every member thinking again alone, asking its best teammate, reading both teammates, and a learned per-member rule, plus a panel mixing the providers that is replayed from stored calls at no cost; the rule and the panel had preregistered tests. (3) Accuracy against cents per question (log scale) on the 144 test questions, each method read once. Open circles: one model alone; squares: the team methods; star: the panel. One GPT-6.1 Sol call at low effort matches every team.

than every member thinking again, with the lower 95% bound of its accuracy difference above −3 points. Neither team's rule passed: Claude −1.4 (−6.3 to +2.8) at $0.022 against $0.044 per question, and GPT +0.7 (−3.5 to +4.9) (CONFIRMATORY). Claude's rule did worse than random choices of the same number of calls (−2.2, −4.8 to −0.1), and GPT's mostly reproduced the plain vote.

Recovering a lost member (PHOENIX; Figure 4). Three agents build a classifier on a public science table (forest cover type, MiniBooNE neutrino events or HIGGS collisions), and halfway through one member is shut down and its model withdrawn. Three recoveries ran under the same caps on dollars, calls, tools and CPU time: ordinary takeover by the next model, a planner model choosing every step, and PHOENIX, which ranks candidate plans by expected gain per dollar along the workflow's dependencies. Every recovered predictor was sealed and then scored once on a held-out split (330 recoveries, balanced-accuracy points). Where the rule can be isolated, with one cheap model doing everything, PHOENIX minus takeover is +0.20 (−0.31 to +0.95; CONFIRMATORY, not resolved). With Claude Opus 5.5 planning it beats takeover by +0.86 (+0.34 to +1.62), but Opus choosing every step without the rule already gains +0.67 (+0.11 to +1.48; POST HOC); the rule adds +0.19 (+0.04 to +0.36) at $0.126 against $0.231 per recovery. With GPT-6.1 Sol planning the gain over takeover is +0.56 (+0.10 to +1.21) and fragile, and two draws of the same method on one case differed by up to 5.85 points. The planner model mattered; the rule's own value is not shown.

8. Choosing members per budget

Because every call is stored, every team in this paper is rescored by replay at no cost, and the best team for each dollar and latency budget can be chosen from calls already made (Figure 5). On the anchor's BBEH questions 42 budget cells have a feasible team, and in every one the winner is a single model: one call in 24 cells and a vote of its own copies in 18. Chosen on half the questions and scored on the other half, the choice loses 0.9–1.6 points on average against the best team for the held-out half (EXPLORATORY). This is stability within one benchmark, not transfer to a new task.

9. Related work

Shared failure and its ceiling. Chen (2026) show that any routing, voting or cascade policy returning one model's answer is bounded by one minus the share of queries every model fails, certify that share from a graded set, and find that a one-factor copula underprices it on open-ended mathemat-

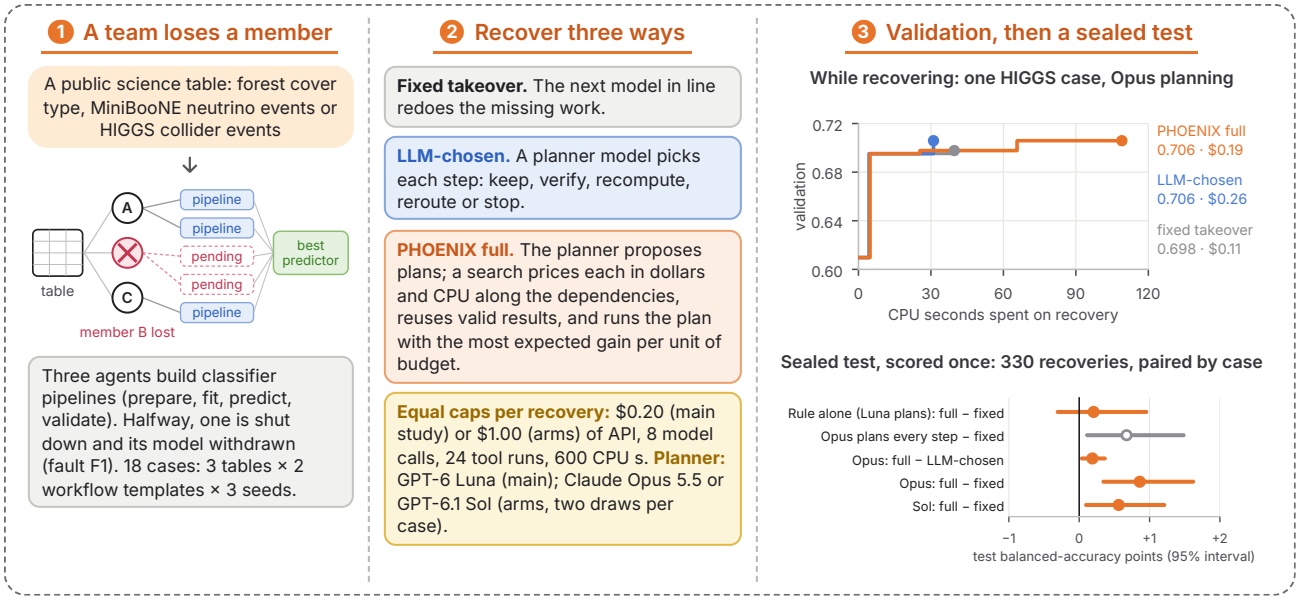


Figure 4. PHOENIX, recovery of a lost team member. (1) Three agents build classifier pipelines on a public science table; halfway, one is shut down and its model withdrawn. (2) Three recoveries run under the same caps per recovery: fixed takeover, a planner model choosing every step, and PHOENIX, which ranks candidate plans by expected gain per unit of budget along the workflow's dependencies. (3) Top: one case's validation score (training-internal rows) as the recovery spends CPU time, with each branch's final score and API dollars. Bottom: the sealed test, scored once, in balanced-accuracy points paired by case (each arm case is the mean of two draws); orange rows are preregistered contrasts, the grey row is post hoc. The rule alone is not resolved; most of the gain over takeover comes from the stronger planner.

ics; they exclude thinking variants and do not hold dollars fixed. We take the ceiling as given and ask how depth moves it and at what price. Kim et al. (2025) measure correlated errors across hundreds of models without an item-difficulty term; Turkmen et al. (2026) prove an error floor under a Gaussian latent factor; Denisov-Blanch et al. (2026) find that polling does not improve truthfulness because errors are shared. Ge et al. (2026) fit an ability-minus-difficulty model to agentic coding leaderboards to predict tasks and agents; we use the same scale for composition.

Width. Chen et al. (2024) show that vote accuracy can rise and then fall with more calls of one model on a mix of easy and hard queries; they count calls, not dollars. Chen et al. (2025) assign different models to the modules of a compound system with a labelled search, which can beat any single model; this is role allocation, not parallel width.

Verification and cascades. Singhi et al. (2025) find self-consistency more compute-efficient than generative verification at most practical budgets, with verification ahead only at many times the compute; our comparator is the judge model's own answer at measured dollars. Conformal cascades (Dou et al., 2026) need labelled calibration data; C3PO (Valkanas et al., 2025) anchors a label-free cascade on the strongest model's answers and so inherits its errors. Learn-then-Test (Angelopoulos et al., 2021) gives the risk control we plan for a runtime router.

10. Limitations

Models and prices are those of October 2026, and public leaderboards mix scaffolds and dates. Most Atlas results are post hoc and uncorrected for the many comparisons made. Contamination cannot be ruled out: BBEH has been public since 2025, so "fresh" means new to this project, and LiveCodeBench predates the testbed's models. The API studies are small (533 items in the anchor) and use two vendors; the AH2 decision covers the first effort step for cheap models on these items only; beyond it the depth study describes flattening gains but does not test depth against width. The anchor's judge test was capped by its pools. The team studies are small (144 LAB-Bench test questions, 18 PHOENIX cases on tables near their accuracy ceiling), several team comparisons are post hoc, and the budget map's stability is measured within one benchmark. Answer format moves the difficulty axis (Chen, 2026); our anchor tasks are free-form. The all-wrong tail on small math sets is not resolved. The anchor's systematic shares have no interval until the gold files are restored.

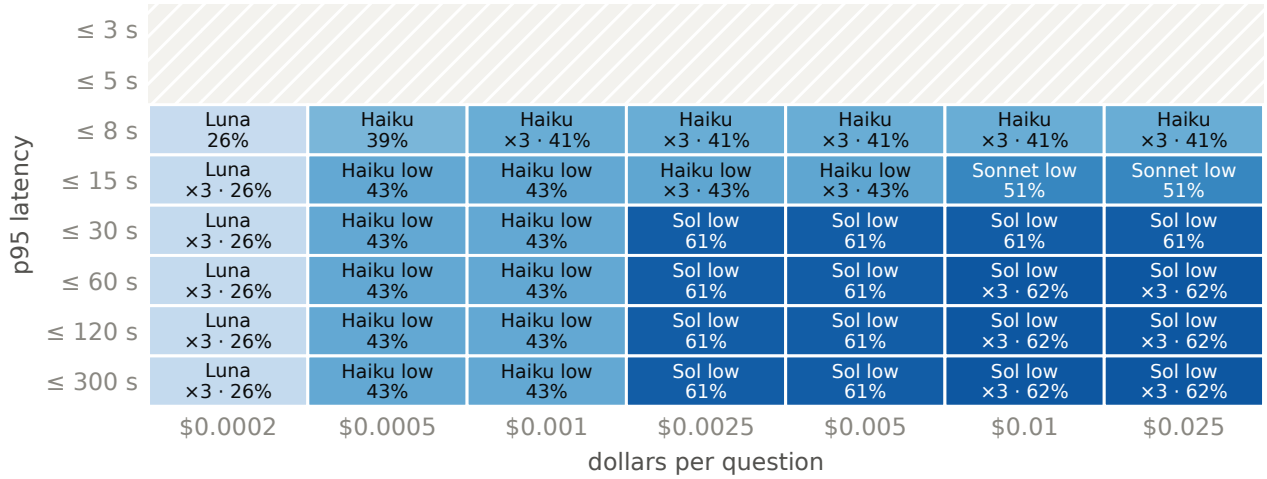


Figure 5. The budget map on the anchor's BBEH questions: the most accurate team within each dollar budget (columns) and 95th-percentile latency budget (rows), chosen by replaying stored calls. Shading is its accuracy; "×3" marks a vote of three calls of one model; hatched rows have no team fast enough. Every winner is a single model.

11. Discussion

The claim in compression terms. A team of copies is an over-parameterized layer with highly correlated channels. Like such a layer it can be pruned heavily at little cost, and like such a layer it gains parameters, not capacity, when it is widened. Its accuracy is bounded by coverage, the share of questions at least one member answers; coverage grows slowly because failures are shared, and a vote or a judge recovers part of the gap between the best member and coverage, never more (Brown et al., 2024; Chen, 2026). The moves that raise capacity are the ones compression predicts: more precision per unit (a step of reasoning) and a larger unit (a stronger member). Coordination, whether votes, critiques, judges or learned rules for when to consult, routes answers among members. It helped where the best member left room, as in the LAB-Bench team of smaller models, but one stronger model reached the same accuracy alone at a fraction of the cost.

Shared failure is measurable, and it says where width pays. One difficulty per question accounts for 0.73 of the latent variation in success on SWE-bench Verified, and a factor model built on it predicts the share of questions every held-out system fails (5.9% predicted, 7.6% observed), where independence and a Gaussian copula do not. The same model says where width does pay: weak systems with different strengths, mixed in one panel (Appendix B). A practitioner can measure the quantity on a small graded pilot: the correlation of the members' errors gives the effective team size, and coverage gives the ceiling of any rule that picks among them.

For practitioners. Spend on depth and on the choice of member before width, and on checking only where the members already cover the answer. Treat agreement among copies as one vote, not many. The budget map makes the choice concrete: on BBEH every winning cell, across dollar and latency budgets, is a single model called once or voted three times, and the choice made on half the questions loses 0.9–1.6 points on the other half (EXPLORATORY).

What remains open. The depth result covers the first reasoning step of cheap models; beyond it gains flatten, and whether depth beats width at the top of a model's range is untested. The budget map is stable within one benchmark; transfer to a new task is the next test. A preregistered handoff test, how much of a strong model's accuracy a cheaper team completes from the strong model's partial reasoning, has run and is not yet read. The next steps (roadmap in research/v4/roadmap.md) are 2,000+ items across 8 or more task families including agentic ones, a third model family used once on the frozen design, the open-weight testbed, and a budget allocator that turns a 30–50 item pilot and a price table into a depth, width and verification mix with predicted accuracy and cost.

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A. Reproducibility

Preregistration. Every paid run carried a committed `prereg:` block (question, prediction, kill rule, decision) that code checked before the run could spend; the full texts are in the repository's research/ folder. Tests whose simulated power at the chosen n fell below 0.70 were reported as exploratory.

Audits. An independent audit read every design and every paid run's outcome before the next step: 135 audits, with 32 GO, 99 ADJUST and 2 STOP verdicts; every ADJUST was resolved before further spend.

Cost. Total API spend $196.64 over 502 runs, under hard per-account caps enforced in code. Every call is stored, so every team in this paper is rescored by replay at no cost. The per-run ledger and the status of every number in the paper are generated in the repository (paper/runs_table.tex, paper/numbers_table.tex).

B. Further results

Votes of one system. A vote of two calls cannot gain in expectation (a right–wrong split is a tie), so gains start at three. The factor model's *vote rule* predicts a k-vote from one-call success p, the share z of failures that abstain (abstentions are non-votes) and the chance q that two wrong answers coincide. Cross-fitted on 695 (system, set) cells with three calls in 24 sets, its mean absolute vote-gain error is 2.0 points against 2.2 for the zero-gain predictor "votes add nothing"; predicted and observed gains correlate at 0.46, and the gain's squared error is smaller by 2.4 points^2 (0.6 to 3.4; item noise adds the same variance to both and cancels). But the rule overstates gains (4.7 predicted, 2.9 observed where it predicts 3 or more), and on the 3-vote accuracy itself it is worse than zero-gain (mean absolute error 4.4 against 4.0 points; squared error larger by -4.2 points^2, -9.1 to -1.6). The rule ranks where votes pay; it does not yet price them.

Votes across systems. Votes pay most when weak systems with different strengths are mixed. On 1,577 panels of three and 1,585 of five MathArena systems with one-call accuracy 0.2–0.7, the plurality of one call each gains 5.7 and 12.5 points over a random member. Cross-fitted, the factor model predicts the panel vote with bias 0.4 and 0.4 points and absolute error 6.4 and 6.6, against 8.2 and 13.6 for zero-gain and 11.9 and 16.2 for independence; it is closer than zero-gain in 4 and 4 of 5 sets, and closer than independence in 4 and 5. Where composition matters, the model prices it without bias; independence overstates it and zero-gain ignores it.

AH1 and AH4 (EXPLORATORY). Sonnet's effort-low setting closed none of the 11.1-point gap to Sol low ($R = 0.02$, -0.32 to 0.26), so the vendor–reasoning confound stays open. The Sol judge's selection efficiency was 0.85 on easy items and 0.14 on hard ones (difference 0.71, 0.41–1.05), the predicted direction, with a ratio whose hard-half denominator is small.

The price of verification on public data. A perfect selector over random 2–5 system SWE-bench sets adds 4.8–8.3 points over the best member, at $0.062–$0.163 of extra spend per point per task. With the anchor's best measured efficiency, a judge over such a set beats the most accurate single submission at no more of its dollars in at most 10% of sets (a perfect selector in at most 13%, judge calls unpriced). On MathArena the room is 1.0–4.5 points because the top systems are close to perfect.

Self-checks and execution. Sol accepted 97% of its own anchor answers (AUROC 0.69); Haiku 66% (AUROC 0.78). On SWE-bench the step count of a run ranks its own successes with median AUROC 0.68; a cascade that escalates on step count never escalates once its threshold is fitted. Execution-based checks on code are part of the open-weight testbed.

The open-weight testbed (design; results pending). The testbed varies precision (BF16, W8, W4, W3), size (Qwen3.5 2B, 4B, 9B) and thinking budget (off, 1k, 4k) with several samples per item on 363 items (200 BBEH, 63 competition math, 100 LiveCodeBench), so width, depth and verification can be bought at measured GPU cost. Its preregistration (T1–T5) has power by simulation of 0.94 (T1: the W4 loss survives voting on hard items), 1.00 (T2), 1.00 (T3: depth against cost-matched width), 1.00 (T4: judge coupling to difficulty) and 0.65 (T5, exploratory: compositional prediction) (DESIGN). The pilot is estimated at $4.1 of GPU time. No GPU has run yet; no result is reported.

C. Additional figures

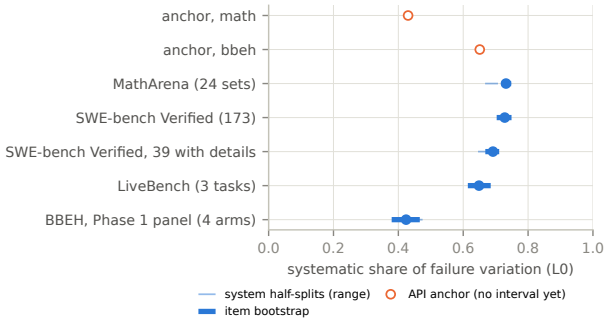


Figure 6. Systematic failure share per source (L0): item-bootstrap 95% interval (thick) and the range over system half-splits (thin; each half fitted on its own). The anchor's shares (hollow) await the gold files for an interval.

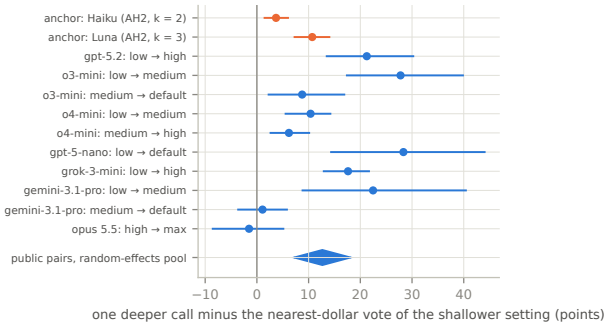


Figure 7. One deeper call minus the nearest-dollar vote of the shallower setting: the anchor's confirmatory AH2 (orange, fresh sealed items, the first effort step of two cheap models) and public MathArena pairs (blue, post hoc) with item-bootstrap 95% intervals and their random-effects pool.

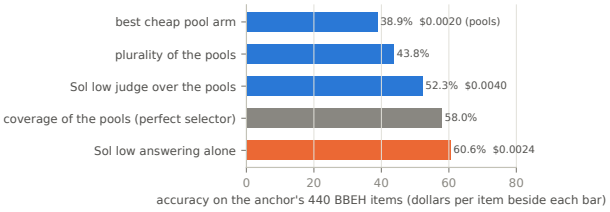


Figure 8. The anchor's BBEH verification read: a judge over cheap pools against the judge model answering alone. The pools' coverage (grey) is below the judge model's own accuracy, which caps every selector over them.

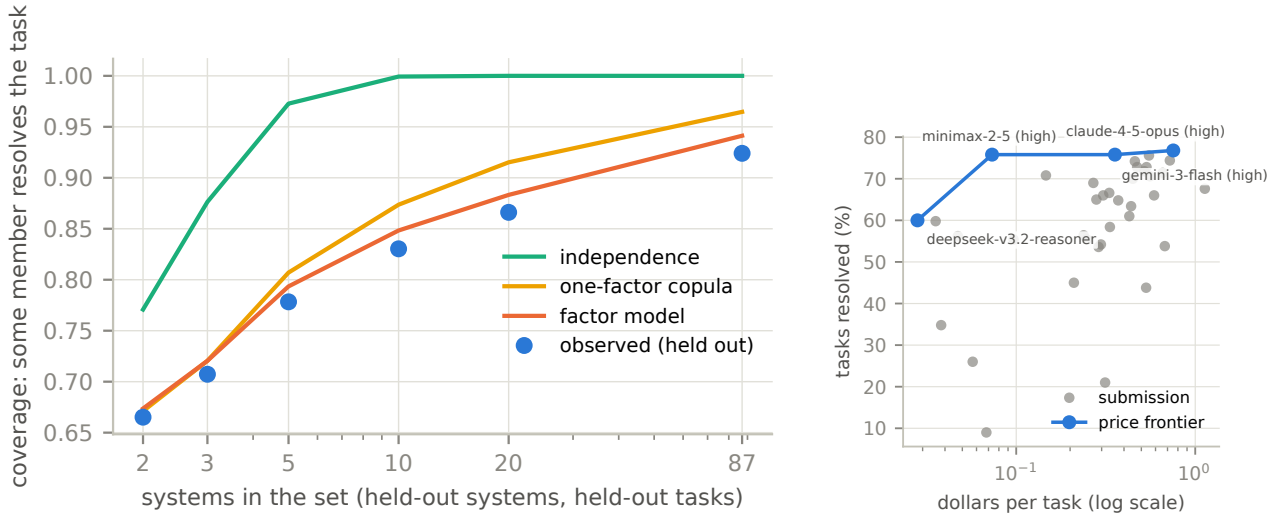


Figure 9. Left: coverage of sets of held-out SWE-bench Verified systems on held-out tasks, observed and predicted from the calibration tasks only (P8). Right: tasks resolved against dollars per task, SWE-bench Verified submissions with costs; the price frontier in blue (A7).

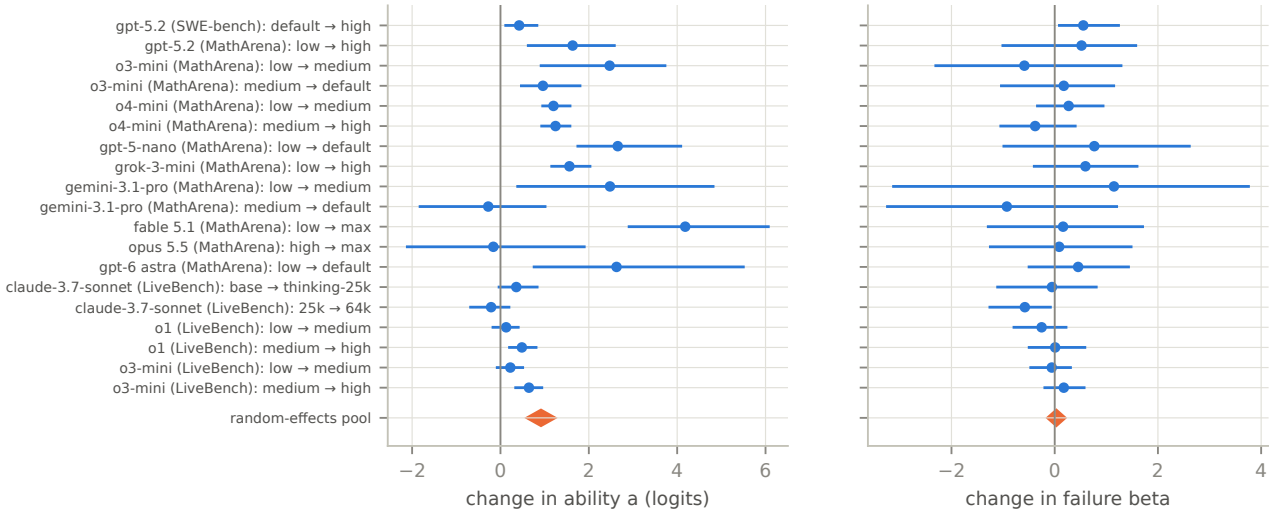


Figure 10. One model at two adjacent reasoning settings (P2): change in ability a and in failure slope β with item-bootstrap 95% intervals, and the DerSimonian–Laird random-effects pool (diamond).

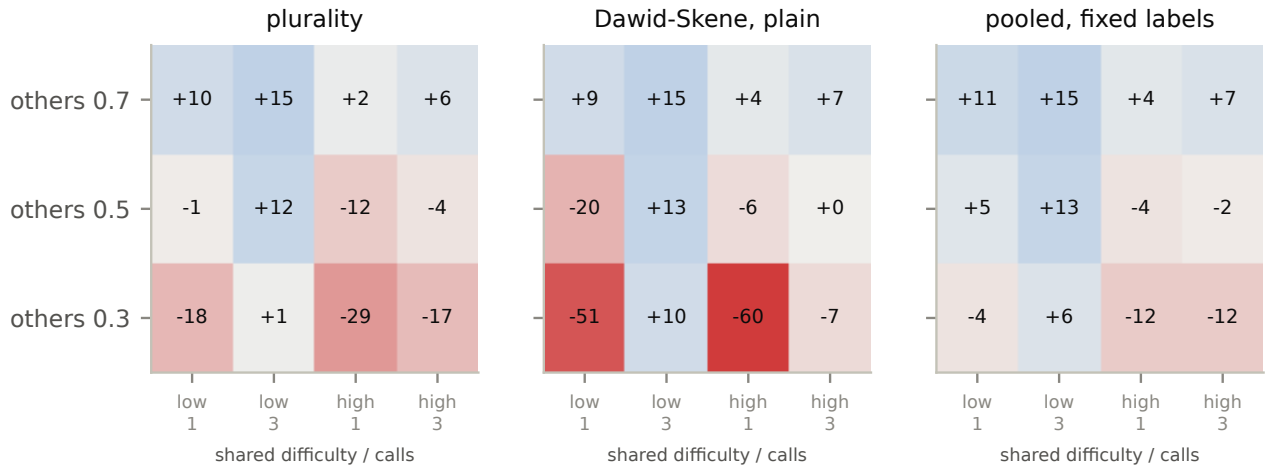


Figure 11. Label-free selection, simulation map (P4): gain over the best system (points) when one system at 0.85 sits among others at 0.3, 0.5 or 0.7, at low or high shared difficulty (systematic share 0.07 or 0.55), with one or three calls per system; averaged over 3, 5 and 9 systems.